

# Disproof of TLMC Conjecture 00000000570

TLMC Falsification Package

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## Abstract

Conjecture 00000000570 asserts that for every acyclic cluster algebra the width of the  $x$ -degree support of the Laurent expansion in the initial cluster (the Chebyshev-normalized invariant) is at most  $2 \cdot \text{rank}$  (“locality of degrees”). We disprove the conjecture with the Kronecker cluster algebra (rank 2, exchange matrix  $B = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$ , affine type  $\tilde{A}_1$ , acyclic): the support width of the cluster variable  $x_6$  is  $6 > 4 = 2 \cdot \text{rank}$ , and the widths  $0, 2, 4, 6, 8, 10, 12, \dots$  of  $x_3, x_4, x_5, \dots$  grow without bound. All numbers are machine-verified: an independent pure-Python recomputation (`reproduce.py`) and a core-Lean 4 formalization with zero axioms and zero `sorry` (`lean4/`).

## 1 The counterexample

Let  $x_1, x_2$  be the initial cluster of the Kronecker cluster algebra (quiver  $1 \rightrightarrows 2$ , acyclic, rank 2). Alternating mutations produce cluster variables satisfying the rank-2 exchange recurrence

$$x_{n+1} x_{n-1} = x_n^2 + 1, \quad x_3 = \frac{x_2^2 + 1}{x_1}.$$

By the Laurent phenomenon each  $x_n$  is a Laurent polynomial in  $x_1, x_2$ ; we compute these expansions exactly (greedy Laurent-polynomial division, verified by remultiplication). Writing the support of  $x_n$  as a set of exponent pairs  $(a, b) \in \mathbb{Z}^2$ , the  $x_1$ -exponent  $a$  spans exactly the interval  $[-(n-2), n-4]$ :

| $n$ | #terms | $x_1$ -range of support | width vs. $2 \cdot \text{rank} = 4$ |
|-----|--------|-------------------------|-------------------------------------|
| 3   | 2      | $[-1, -1]$              | 0 (ok)                              |
| 4   | 4      | $[-2, 0]$               | 2 (ok)                              |
| 5   | 7      | $[-3, 1]$               | 4 (ok)                              |
| 6   | 11     | $[-4, 2]$               | <b>6 &gt; 4 (violation)</b>         |
| 7   | 16     | $[-5, 3]$               | 8 > 4                               |
| 8   | 22     | $[-6, 4]$               | 10 > 4                              |
| 9   | 29     | $[-7, 5]$               | 12 > 4                              |

Table 1: Support widths along the Kronecker chain. The values  $0, 2, 4, 6, 8, 10, 12$  for  $x_3, \dots, x_9$  match the recorded attack exactly.

Both endpoints of the strip are attained: for  $x_6$ ,

$$x_6 = x_1^{-4}x_2^{-3} + 4x_1^{-4}x_2^{-1} + 6x_1^{-4}x_2 + 4x_1^{-4}x_2^3 + x_1^{-4}x_2^5 + 3x_1^{-2}x_2^{-3} + 6x_1^{-2}x_2^{-1} + 3x_1^{-2}x_2 + 3x_2^{-3} + 2x_2^{-1} + x_1^2x_2^{-3},$$

with the corner coefficients equal to 1, so the width is exactly  $6 > 4$ . The growth is unbounded:  $\text{width}(x_n) = 2(n-3)$ ; the accompanying script verifies this to  $n = 25$  (width 44). The inductive

mechanism is that the leading  $x_1$ -exponent of  $x_{n+1} = (x_n^2 + 1)/x_{n-1}$  is twice the top exponent of  $x_n$  minus the bottom exponent of  $x_{n-1}$ , widening the interval by one on each side per step. All other natural spreads ( $x_2$ -exponent, total degree  $x_1x_2$ , and  $x_1x_2^{-1}$ -degree) also grow by 2 per step, so the disproof does not depend on the width convention.

## 2 Why the numbers are the cluster variables

The script asserts the seven Laurent-polynomial identities  $x_{n-1}x_{n+1} = x_n^2 + 1$  ( $n = 2, \dots, 8$ ) exactly, and every division is verified by remultiplication. Since  $\mathbb{Z}[x_1^{\pm 1}, x_2^{\pm 1}]$  is an integral domain, the Laurent polynomial  $q$  with  $q \cdot x_{n-1} = x_n^2 + 1$  is unique; hence the verified tables *are* the Laurent expansions of the cluster variables  $x_1, \dots, x_9$ . The Kronecker quiver is acyclic and stays acyclic under mutation, so the counterexample lies inside the conjecture’s hypothesis, and  $\text{width}(x_6) = 6 > 2 \cdot \text{rank} = 4$  falsifies it.

## 3 Machine verification

**Lean 4** (`lean4/`, core Lean only, no Mathlib): the tables TBL1–TBL9 are verified against all seven exchange relations coefficientwise (`relz2–relz8`), the strip bounds and attained endpoints give the exact widths 0, 2, 4, 6, 8, 10, 12 (`width3–width9`), and `disproof_00000000570` packages the violations  $\text{width}(x_6) = 6 > 4$  and  $\text{width}(x_9) = 12 > 4$ . `Check.lean` audits `#print axioms` for every declaration: all report “does not depend on any axioms” (zero axioms, zero `sorry`; `decide` only on closed propositions).

**Python** (`reproduce.py`, no dependencies): recomputes  $x_3, \dots, x_{25}$  from scratch, verifies all divisions and the seven exchange identities, prints the width table and the  $b = 1$  boundary comparison, and emits the Lean tables (byte-identical to `lean4/Main.lean`).

## 4 Boundary

- *Rank 1*: the width is  $0 \leq 2 \cdot \text{rank}$ ; holds.
- *Rank 2, finite type  $A_2$  ( $b = 1$ )*: widths stay  $\leq 1 \leq 4$ ; holds.
- *Rank 2, affine  $\tilde{A}_1$  ( $b = 2$ , this attack)*: fails from  $x_6$  on, unboundedly.
- *Rank  $R \geq 3$  acyclic*: the product Kronecker  $\times A_1^{R-2}$  is acyclic of rank  $R$  with the same support widths, so the bound fails for every  $R \geq 2$ .
- *Chebyshev normalization*: normalizing cluster variables by Laurent monomials translates supports uniformly and leaves every width invariant, so the attack survives the normalization.